Richard Haburcak

Curriculum Vitae

Education

2018–2023 **Ph.D. in Mathematics**, Dartmouth College, Hanover, New Hampshire
Advisor: Asher Auel
Thesis: Brill–Noether Theory via K3 Surfaces

2014–2018 **M.A. in Chemistry**, Brandeis University, Waltham, Massachusetts
Advisor: Bing Xu
Thesis: Ligand-Receptor Interactions Modulate Self-Assembly of Small Molecules

2014–2018 **B.A. in Mathematics and Chemistry**, Brandeis University, Waltham, Massachusetts

Appointments

2024-Present **RTG Visiting Assistant Professor**, The Ohio State University, Columbus, Ohio

Spring 2024 **Postdoctoral Fellow**, Max Planck Institute for Mathematics, Bonn, Germany

Fall 2023 **Junior Trimester Program**, Hausdorff Research Institute for Mathematics, Bonn, Germany

2023-2024 **Visiting Scholar**, Dartmouth College, Hanover, New Hampshire

Awards and Grants

2025-2027 **AMS-Simons Travel Grant**, *$6,000*

2023 **Outstanding Graduate Teacher**, *Dartmouth College*, student nominated award

2018 **Shapiro Prize in Mathematics**, *Brandeis University*

2017 **Goldwater Scholarship Honorable Mention**

2016 **Science Hall of Fame**, *Brandeis University*

Publications and Preprints

Preprints

Link *Brill–Noether loci in genus* ≤ 12, Richard Haburcak. 2025. Submitted.

Link *Distinguishing Brill–Noether loci*, Asher Auel, Richard Haburcak, and Andreas Leopold Knutsen. 2024. Submitted.

Publications

Link *Some reducible and irreducible Brill–Noether loci*, Richard Haburcak and Montserrat Teixidor i Bigas. Math. Z. **312**, 102 (2026).

Link *Maximal Brill–Noether loci via K3 surfaces*, Asher Auel and Richard Haburcak. International Mathematics Research Notices (IMRN), **2025**(20), 2025.

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Link *Maximal Brill–Noether loci via degenerations and double covers*, Andrei Bud and Richard Haburcak. 2024. Accepted to Ann. Sc. Norm. Super. Pisa Cl. Sci. (5).

Link *Holomorphic forms and non-tautological cycles on moduli spaces of curves*, Veronica Arena, Samir Canning, Emily Clader, Richard Haburcak, Amy Q. Li, Siao Chi Mok, and Carolina Tamborini. Sel. Math. New Ser. 31, 47 (2025).

Link *Maximal Brill–Noether loci via the gonality stratification*, Asher Auel, Richard Haburcak, and Hannah Larson. Accepted to Michigan Math. J.

Link *Curves on Brill–Noether special K3 surfaces*, Richard Haburcak. Math. Nachr. 297 (2024), 4479–4509.

Previous Chemistry Work

Link *Ligand–Receptor Interaction Modulates the Energy Landscape of Enzyme-Instructed Self-Assembly of Small Molecules*, Richard Haburcak , Junfeng Shi, Xuewen Du, Dan Yuan, and Bing Xu*, *J. Am. Chem. Soc.* **2016**, 138, 47, 15397–15404.

Link *Nanonets Collect Cancer Secretome from Pericellular Space*, Rong Zhou, Yi Kuang, Jie Zhou, Xuewen Du, Jie Li, Junfeng Shi, Richard Haburcak, Bing Xu*, *PLOS One* **2016**, 11, e0154126.

Link *Enzymatic Dissolution of Biocomposite Solids Consisting of Phosphopeptides to Form Supramolecular Hydrogels*, Junfeng Shi, Dan Yuan, Richard Haburcak, Qiang Zhang, Dr. Chao Zhao, Prof. Dr. Xixiang Zhang, PRof. Dr. Bing Xu*, *Chem. Eur. J.* **2015**, 21, 18047-18051

Link *Supramolecular Detoxification of Neurotoxic Nanofibrils of Small Molecules via Morphological Switch*, Junfeng Shi, Xuewen Du, Dan Yuan, Richard Haburcak, Ning Zhou, and Bing Xu*, *Bioconjugate Chem.* **2015**, 26, 9, 1879–1883.

Link *Enzyme transformation to modulate the ligand–receptor interactions between small molecules*, Junfeng Shi, Xuewen Du, Dan Yuan, Richard Haburcak, Dongdong Wu, Ning Zhou, and Bing Xu*, *Chem. Commun.* **2015**, 51, 4899-4901

Link *Ligand–Receptor Interaction Catalyzes the Aggregation of Small Molecules To Induce Cell Necroptosis*, Junfeng Shi, Xuewen Du, Yibing Huang, Jie Zhou, Dan Yuan, Dongdong Wu, Ye Zhang, Richard Haburcak, Irving R. Epstein*, and Bing Xu*, *J. Am. Chem. Soc.* **2015**, 137, 26–29.

Teaching experience

The Ohio State University, Instructor of record

Spring 2026 **Abstract Algebra 1**, *Math 4580*

Fall 2025 **Abstract Algebra 2**, *Math 4581*

Spring 2025 **Introduction to ODEs**, *Math 2255*

Fall 2024 **Introduction to ODEs**, *Math 2255*

Dartmouth College, Instructor of Record

Winter 2023 **Multivariable Calculus**, *Math 13*

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Winter 2022 **Calculus**, *Math 3*

Fall 2020 **Introduction to Calculus**, *Math 1*

Teaching Assistant

Winter 2020 **Graduate Algebra**, *Math 101*, Dartmouth College

Fall 2019 **Calculus of Functions of one and Several Variables**, *Math 8*, Dartmouth College

Spring 2019 **Differential Equations**, *Math 23*, Dartmouth College

Fall 2018 **Advanced Multivariable Calculus**, *Math 13*, Dartmouth College

2015–2017 **Quantum Chemistry**, *Chem 142*, Brandeis University, 3 times

2017–2018 **Thermodynamics**, *Chem 141*, Brandeis University, 2 times

Talks

Invited Conference Talks

April 2026 **K3 surfaces and Brill–Noether theory**, *AMS Spring Central Sectional on Classical questions through the lens of birational algebraic geometry*, North Dakota State University, Fargo, North Dakota

Jan 2026 **Refined Brill–Noether theory**, *AMS Special Session on Combinatorics and moduli spaces*, JMM 2026, Washington, D.C.

Oct 2025 **Brill–Noether loci**, *Boston Algebraic Geometry Day*, Tufts University, Medford, Massachusetts

June 2025 **Limit linear series, theta characteristics, and Brill–Noether loci**, *3rd Workshop of the Kentucky–Ohio ALgebra Alliance (KOALA 2025)*, The Ohio state University, Columbus, Ohio

Jan 2024 **The Maximum Gonality in a Brill–Noether Locus**, *AMS Special Session on Combinatorial Perspectives on Algebraic Curves and their Moduli*, JMM 2024, San Francisco, California

April 2023 **Brill–Noether special cubic fourfolds of discriminant** 26, *AMS Spring Central Sectional on Brauer Groups in Algebraic Geometry and Arithmetic*, University of Cincinnati, Cincinnati, Ohio

Oct 2022 **Brill–Noether Special K3 Surfaces**, *AMS Fall Southeastern Sectional on Geometry and Arithmetic of Hyperkähler Manifolds*, University of Tennessee, Chattanooga, Tennessee

Invited Seminar Talks

Oct 2025 **Brill–Noether loci in low genus**, *Algebra Seminar*, University of Kentucky, Lexington, Kentucky

Sept 2025 **Special Algebraic Curves**, *Ohio State University Math Welcome Seminar*, The Ohio State University, Columbus, Ohio

March 2025 **Brill–Noether loci in low genus**, *Algebra, Geometry, and Number Theory Seminar*, Tufts University, Medford, Massachusetts

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March 2025 **Brill–Noether loci in low genus**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2025 **Non-tautological Cycles on Moduli Spaces of Smooth Curves**, *Arithmetic Geometry Seminar*, The Ohio State University, Columbus, Ohio

Aug 2024 **Distinguishing Brill–Noether loci**, *Algebraic Geometry Seminar*, The Ohio State University, Columbus, Ohio

June 2024 **Distinguishing Brill–Noether loci**, *Seminar Algebraic Geometry*, MPIM Bonn, Bonn, Germany

May 2024 **Maximal Brill–Noether loci**, *Oberseminar*, MPIM Bonn, Bonn, Germany

April 2024 **(Algebraic) Surfaces, lattice polarized K3 surfaces, and Brill–Noether special K3s**, *Geometry and Topology Seminar*, University of Iowa, Iowa City, Iowa

Dec 2023 **Maximal Brill–Noether loci**, *Geometry Seminar*, Università Roma Tre, Rome, Italy

Nov 2023 **Maximal Brill–Noether loci**, *Research Seminar Algebraic Geometry*, Leibniz Universität Hannover, Hannover, Germany

Oct 2022 **Maximal Brill–Noether loci via K3 Surfaces**, *Algebra Seminar*, University of Kentucky, Lexington, Kentucky

Sept 2022 **Maximal Brill–Noether loci via K3 Surfaces**, *Algebra Seminar*, Brown University, Providence, Rhode Island

April 2022 **Maximal Brill–Noether Loci via Lifting Line Bundles**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

March 2021 **Brill-Noether Special K3 Surfaces**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Expository and Reading Seminar Talks

March 2026 **(Moduli stack of) Vector Bundles on P1 (cont.)**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

Feb 2026 **(Moduli stack of) Vector bundles on P1**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

Oct 2025 **Degeneracy Loci**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

Nov 2023 **Torelli theorems for cubic threefolds**, *STReTCH seminar*, Hausdorff Institute for Mathematics, Bonn, Germany

April 2023 **Why are word problems hard? (Introduction to the word problem for groups)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Oct 2022 **Galois Cohomology and Local Class Field Theory**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Oct 2022 **All Triangles Form a Triangle**, *Undergraduate Math Club*, University of Kentucky, Lexington, Kentucky

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Sept 2022 **Triangles. Boundaries. And...? (Introduction to moduli spaces of triangles)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

July 2022 **Ètale Cohomology**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

July 2022 **Group and Galois Cohomology**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

April 2022 **Sheaves on Graphs**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2022 **Introduction to Algebraic Surfaces I**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Oct 2021 **Categories, Particle Physics, and Yoneda**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

May 2021 **Finite groups, étale groups, Cartier duality**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

April 2021 **Projective Space and Lines**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

March 2021 **Nekovář's p-adic Heights**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Jan 2021 **Sheafy Vector Bundles and Chern Classes**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Nov 2020 **Multiplying Divisors**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2020 **Maximal Spectra and Why Continuous Functions are Important**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Sept 2019 **Semisimplicity of Finite Group Rings**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

April 2019 **Adding and Putting Rings on It (Abelian Categories)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Service and Outreach

Fall 2025-26 **Mentor**, *OSU Cycle (undergraduate reading and research)*, The Ohio State University, Columbus, Ohio

Spring 2023 **Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Winter 2023 **Co-organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Spring 2022 **Organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Winter 2022 **Organizer**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

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Winter 2021 **Organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Spring 2019 **Co-organizer**, *Math table at Dartmouth Science Day*, Dartmouth College, Hanover, New Hampshire

Conference Organizing

Spring 2026 **Co-organizer**, *AMS Special Session on Moduli and Syzygies of Algebraic Curves, as part of AMS Spring Eastern Sectional*, Boston College, Boston, Massachusetts

Fall 2023 **Co-organizer**, *School on K3 surfaces, hyperkähler manifolds, and cubic fourfolds, as part of a Junior Trimester Program*, Hausdorff Research Institute for Mathematics, Bonn, Germany

Seminar Organizing

2025-Present **Co-organizer**, *Ohio State University Algebraic Geometry Seminar*, The Ohio State University, Columbus, Ohio

Summer 2022 **Co-organizer**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

2021-2022 **Co-organizer**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Conferences and Workshops Attended

April 2026 **AMS Spring Central Sectional**, North Dakota State University, Fargo, North Dakota

Mar 2026 **AMS Spring Eastern Sectional**, Boston College,Boston, Massachusetts

Jan 2026 **VBAC 2026**, Universidad Autónoma de Yucatán (UADY), Mérida, Yucatán, Mexico

Jan 2026 **JMM 2026**, Washington, D.C.

Nov 2025 **Algebra, Geometry and Combinatorics Day (AlGeCom) XXVI**, The Ohio State University, Columbus, Ohio

Oct 2025 **Boston Algebraic Geometry Day** , Tufts University, Medford, Massachusetts

Sept 2025 **Workshop on Combinatorics and Geometry of Moduli of Curves**, University of Illinois Urbana-Champaign, Champaign, Illinois

July 2025 **Summer Research Institute in Algebraic Geometry**, Colorado State, Fort Collins, Colorado

July 2025 **Bootcamp preceding Summer Research Institute in Algebraic Geometry**, Colorado State, Fort Collins, Colorado

June 2025 **3rd Workshop of the Kentucky-Ohio ALgebra Alliance (KOALA 2025)**, The Ohio State University, Columbus, Ohio

May 2025 **Arithmetic, K-theory, and Algebraic Cycles**, The Ohio State University, Columbus, Ohio

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March 2025	**MAGNTS 2025**, The Ohio State University, Columbus, Ohio
March 2025	**Georgia Algebraic Geometry Symposium 2025**, Univerisyt of Georgia, Athena, Georgia
Nov 2024	**AGNES 2024**, Dartmouth College, Hanover, New Hampshire
Feb 2024	**A panorama of moduli spaces**, Goethe University Frankfurt am Main, Frankfurt, Germany
Jan 2024	**JMM 2024**, San Francisco, California
Oct 2023	**Vector bundles and combinatorial algebraic geometry**, Goethe University Frankfurt am Main, Frankfurt, Germany
Sept 2023	**School on K3 surfaces, hyperkähler manifolds, and cubic fourfolds**, Hausdorff Research Institute for Mathematics, Bonn, Germany
Aug 2023	**VBAC 2023**, Universität Duisburg-Essen, Essen, Germany
July 2023	**AGNES Summer School on Intersection Theory on Moduli Spaces**, Brown University, Providence, Rhode Island
July 2023	**GAeL XXX**, University of Warwick, UK
April 2023	**AGNES 2023**, Stony Brook University, Stony Brook, New York
April 2023	**AMS Spring Central Sectional**, University of Cincinnati, Cincinnati, Ohio
Jan 2023	**JMM 2023**, Boston, Massachusetts
Dec 2022	**Birational Complexity of Algebraic Varieties 2022**, Simons Center, Stony Brook, New York
Dec 2022	**RTG/SCGP Workshop on Birational Complexity of Algebraic Varieties 2022**, Simons Center, Stony Brook, New York
Nov 2022	**AGNES 2022**, UMass Amherst, Amherst, Massachusetts
Oct 2022	**AMS Fall Southeastern Sectional**, University of Tennessee, Chattanooga, Tennessee
Oct 2022	**AMS Fall Eastern Sectional**, UMass Amherst, Amherst, Massachusetts
Sept 2022	**BATMOBILE 2022**, Amherst College, Amherst, Massachusetts
Aug 2022	**AGNES Summer School on Higher Dimensional Moduli**, Brown University, Providence, Rhode Island
July 2022	**VBAC 2022**, University of Warwick, UK
May 2022	**AGNES 2022**, Rutgers University, New Brunswick, New Jersey
May 2022	**BATMOBILE 2022**, Brown University, Providence, Rhode Island
Oct 2021	**AGNES 2021**, Boston College, virtual
Jan 2021	**JMM 2021**, virtual
Oct 2020	**AGNES 2020**, Stony Brook, virtual

Referee

Semigroup Forum

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Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg
Algebraic Geometry
Mathematische Zeitschrift

Press

Math Directed Reading Program Fosters Mentorship and Inclusion

Memberships

American Mathematical Society (AMS).

Languages

English: native speaker. **Slovak:** fluent. **Spanish:** intermediate.

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